Web Science

Assessed Coursework

Reddit Data Analysis

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Introduction

Online communities are dynamic ecosystems where user interactions shape the flow of information, influence discussions, and contribute to overall community cohesion. In this study, we examine the r/InvestmentClub subreddit to understand the role **super users** play in in maintaining the structure and engagement of the community.

Using **network analysis**, we construct and analyse interaction graphs to measure user connectivity and engagement patterns. The main research focus is to determine **how central super users are to the cohesiveness and functioning of the community**, using key network metrics such as degree centrality, rich-club coefficient, and Z-score analysis. Beyond network structure, we analyse trends over time**,** paying special attention to a noticeable spike in user interactions after 2020 and aim to understand the driving forces behind this shift.

By combining graph theory and topic modeling, this study provides a comprehensive view of user influence, content evolution, and the broader behavioral patterns within the subreddit.

2. Data Cleaning and Preprocessing

The raw datasets of submissions and comments from the subreddit contained many columns that were irrelevant to the research objectives. To ensure the focus remained on the key aspects of user behavior, we carefully selected the most pertinent features from each dataset. The data cleaning process included the removal of unnecessary columns and rows as well as standardisation and transformation of certain data points.

2.1 Submissions Dataset

The columns retained in the submissions dataset are as follows:

- **id**: Unique identifier for each post.
- **author**: The user who posted the submission.
- **title**: The title of the post.
- **created_utc**: Timestamp for when the submission was created.
- **score**: The score of the submission.
- **num_comments**: The number of comments on the post.

These columns were selected because they capture the essential characteristics of submissions, such as the content (title), engagement (score, num_comments), and temporal data (created_utc) which are relevant for analysing user interaction patterns and temporal trends.

2.2 Comments Dataset

The columns retained in the comments dataset are as follows:

- **id**: Unique identifier for each comment.
- **author**: The username of the person posting the comment.
- **parent_id**: The identifier for the parent submission or comment.
- **unique_id**: A modified version of the parent_id created to make connections between comments and submissions.
- **body**: The text content of the comment.
- **created_utc**: Timestamp for when the comment was posted.
- **score**: The score of the comment (number of upvotes minus downvotes).
- **link_id**: The identifier for the submission the comment belongs to.
- **ups**: The number of upvotes the comment received.

- **downs**: The number of downvotes the comment received.

These columns were essential for analysing user interactions and constructing the network graph based on comment behavior and capturing the sentiment and engagement level of comments.

2.3 Data Cleaning:

1. **Column Renaming and Standardization**: We standardised column names to maintain uniformity across both datasets. For example, we ensured that the parent_id column in the comments dataset was consistent in formatting.
2. **Handling Missing Values**: The comments dataset did not contain any missing values, and similarly, the submissions dataset had complete data for all columns we kept.
3. **Creating the unique_id**: To facilitate the merging of the two datasets, we created a unique_id in the comments dataset by removing the prefixes "t3_" and "t1_" from the parent_id column in df2. This unique_id allowed us to connect comments to their parent posts ensuring a consistent identifier for merging.
4. **Datetime Conversion**: The created_utc columns in both datasets were converted to a standard datetime format using pd.to_datetime() ensuring uniformity in the representation of timestamps.

2.4 Removed Columns

Several columns were excluded from the final dataset as they did not directly contribute to understanding user behavior and community dynamics.

1. **Excessive Missing Values**:
   Columns like author_flair_css_class, approved_at_utc, removal_reason, and report_reasons contained significant portions of missing data or had no entries at all. These would not contribute useful information and could introduce noise.
2. **Low Relevance to Research Focus**:
   Features such as can_gild, is_video, and no_follow were related to Reddit’s platform functionality or content metadata but did not provide valuable insights.
3. **Low Variance**:
   Features such as score_hidden, likes, and removed_by had very few non-null values or

were dominated by a single value. These were removed as they lacked variability and did not contribute to meaningful analysis.

2.5 Post-Merge Transformation

After cleaning, the comments dataset was filtered to include only rows where the unique_id existed in the id column of the submissions dataset. The two datasets were then merged using an inner join on the unique_id (from the comments dataset) and id (from the submissions dataset). The inner join was chosen to ensure that only the records that had matching comments and submissions were included in the final dataset. This approach eliminates any comments without corresponding submissions and vice versa, ensuring the integrity and relevance of the merged data for further analysis.

By focusing on the key rows and features, the dataset was made cleaner, more efficient, and directly aligned with the research objectives.

3. Network Graph Construction

3.1 Building the Graph

For building the graph, we utilise the merged dataset that combines both comments and submissions data. The key elements of the graph are:

- **Origin Nodes**: In this context, the "origin" nodes represent the users who make the initial comment, i.e., the users whose posts or comments initiate the interaction. These are the nodes identified as "author_comment" in the dataset.
- **Destination Nodes**: The "destination" nodes are the users who reply to the initial comment, which are the "unique_id" values in the dataset. These represent the users who engage with the comment by replying.

We use a directed graph (DiGraph) to represent the relationships between the author and the replied user. Each interaction (comment-reply relationship) is modeled as a directed edge from the origin node (commenter) to the destination node (replying user). This allows us to study the flow of interactions in the community.

The approach captures the dynamic nature of interactions in a discussion forum where one user's comment might lead to multiple replies forming a directed acyclic graph that reflects the interaction hierarchy. The directed edges allow us to analyse the directionality of communication and how different users connect and influence one another.

Pseudocode for Building the Graph:

<u>Create a graph (G):</u>
- Import necessary libraries:
  - import networkx as nx
- Initialize the directed graph (or undirected graph depending on your data):
  - G = nx.DiGraph()

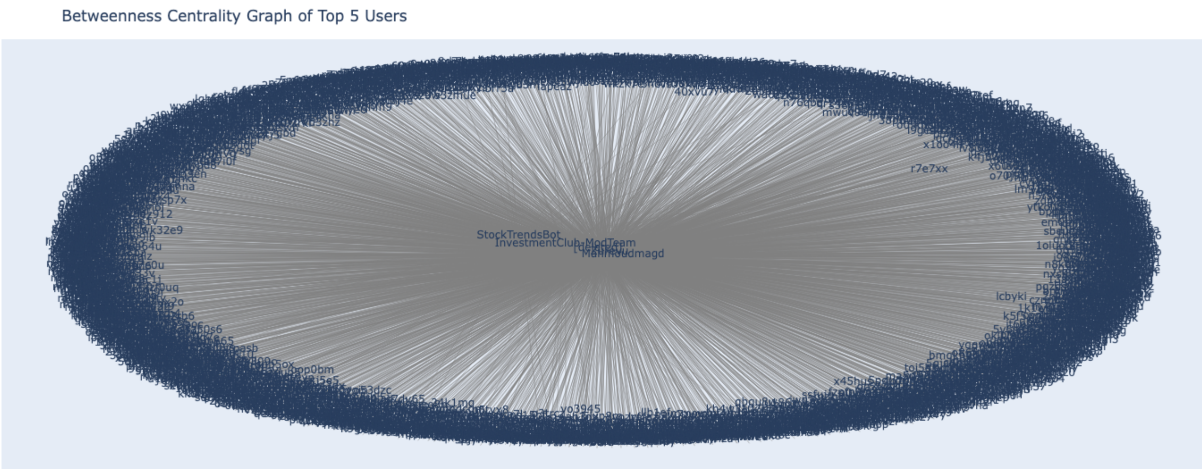
<u>Add nodes and edges to the graph:</u>
- For each user in the dataset, add them as a node:
  - G.add_node(user)
- For each interaction or relationship between users, add an edge:
  - G.add_edge(user, reply_to)

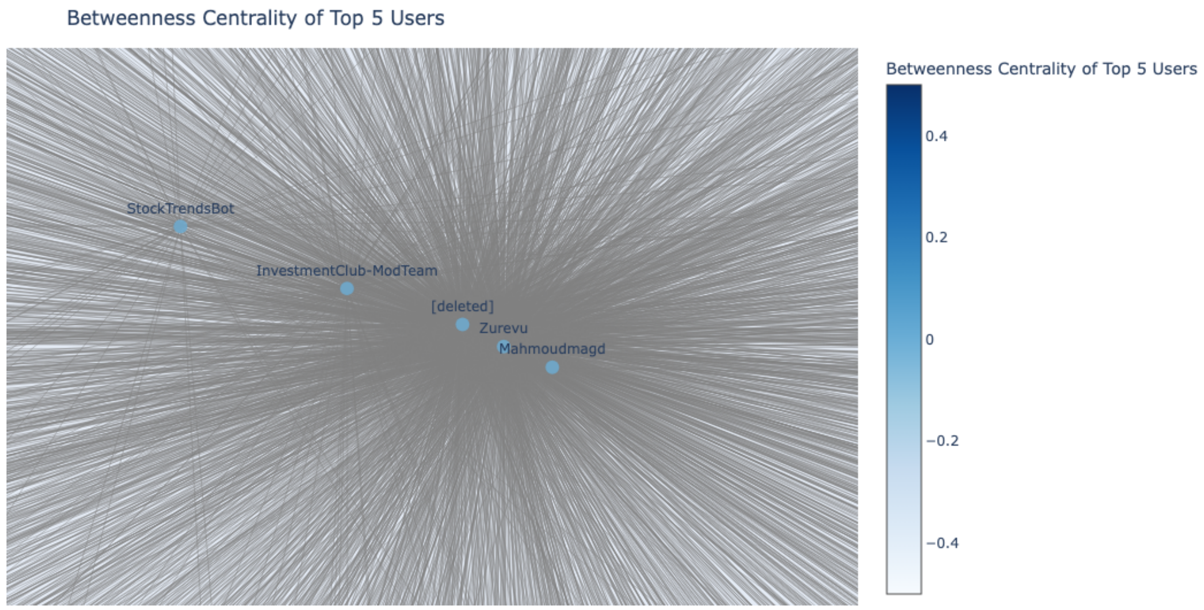
3.2 Graph Visualisations

In the analysis, we created four key visualizations, each representing different aspects of user interactions and centrality measures. These visualizations help us understand the structure and dynamics of interactions within the subreddit, specifically focusing on the top 5 most active users.

- **Betweenness Centrality**

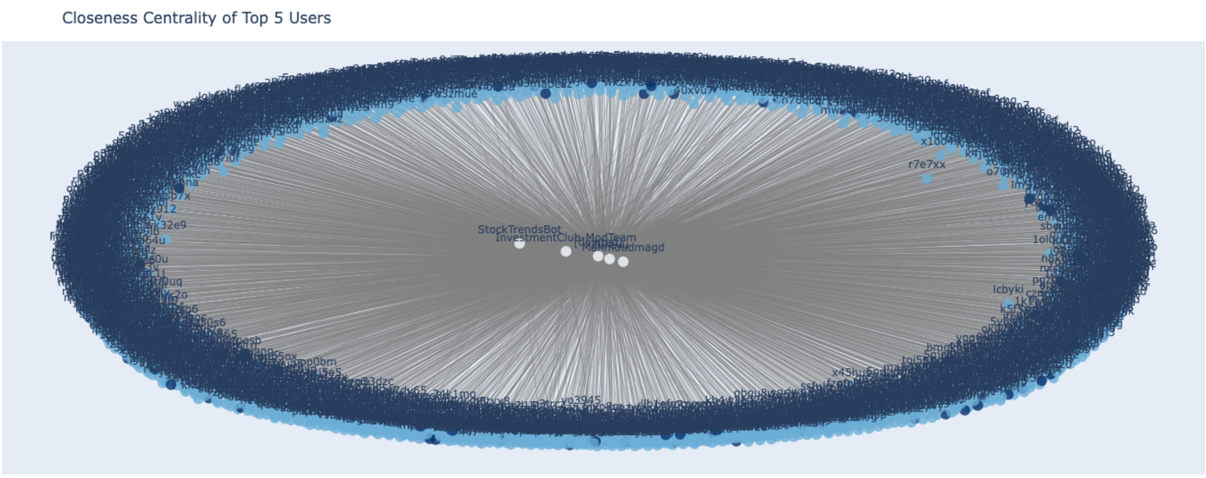
Betweenness centrality measures how often a node acts as a bridge along the shortest paths between other nodes in a network. Nodes with higher betweenness centrality tend to be more centrally located, meaning they connect different groups and facilitate information flow. It is calculated by counting how many shortest paths pass through a node relative to all shortest paths in the network. High-betweenness nodes are often key intermediaries and their removal can significantly impact network connectivity.

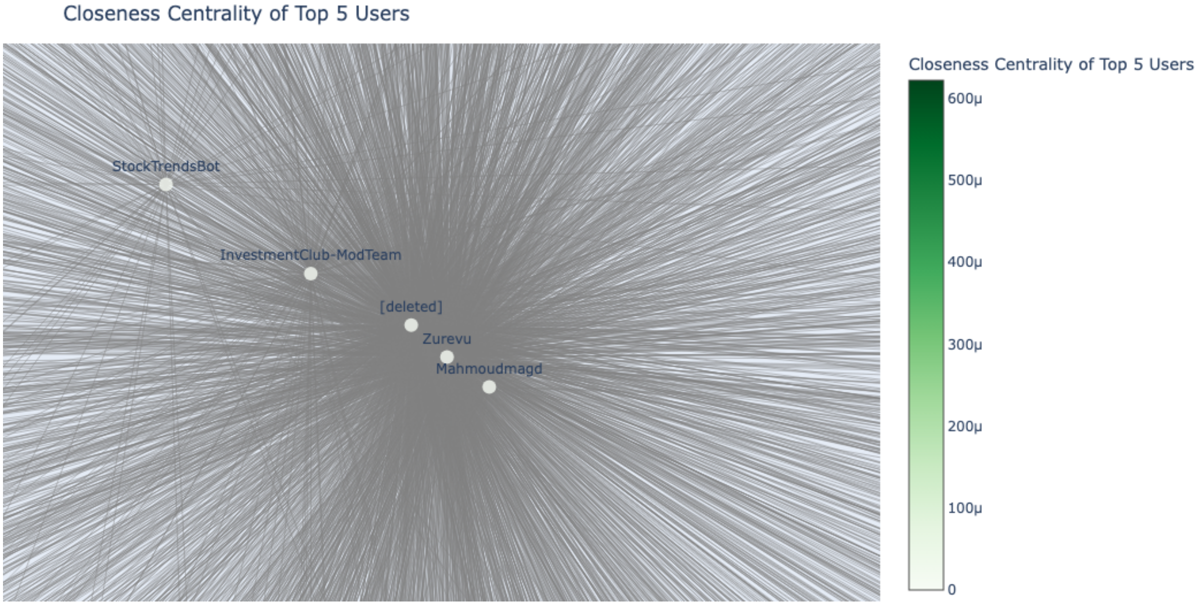




- **Closeness Centrality:**

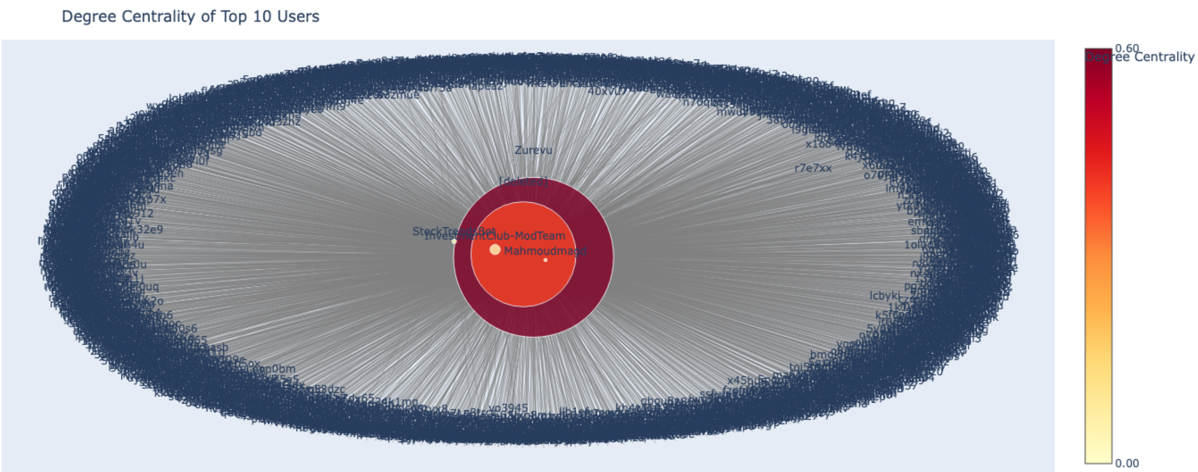
Closeness centrality measures how efficiently a node is connected to all other nodes in a network. Nodes with higher closeness centrality are more centrally located, meaning they have shorter average distances to all other nodes. It is calculated as the inverse of the average shortest path length from a node to every other node in the network. Nodes with high closeness centrality can quickly reach or spread information across the network, making them influential in communication and diffusion processes.

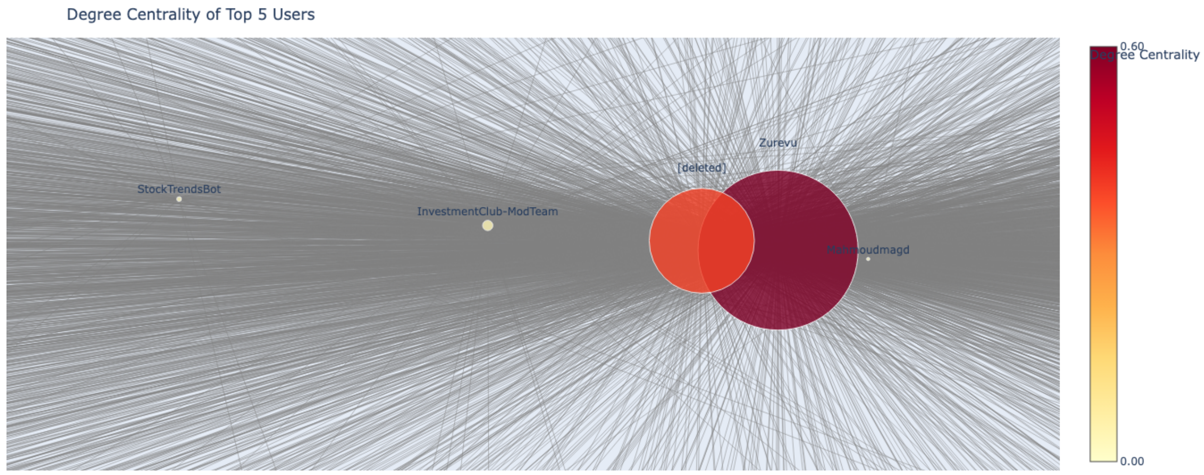




- **Degree Centrality:**

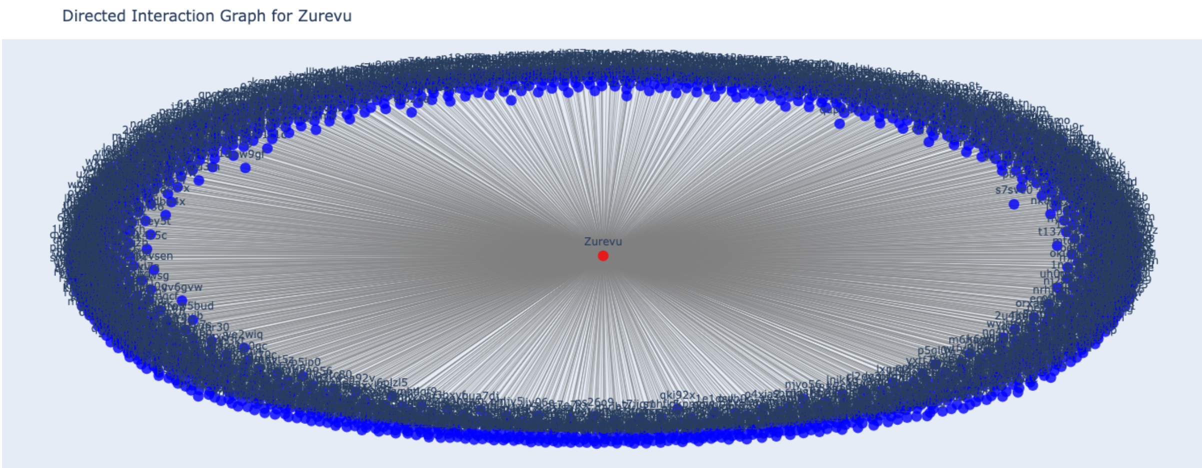
Degree centrality measures the number of direct connections a node has in a network. This visualization shows the degree centrality of the top 5 users with the size of each node representing the number of direct connections a user has. The color of each node reflects the degree centrality, with darker shades indicating higher centrality, highlighting users who are more directly connected within the network. Users with higher degree centrality are often considered more influential due to their greater number of direct interactions.

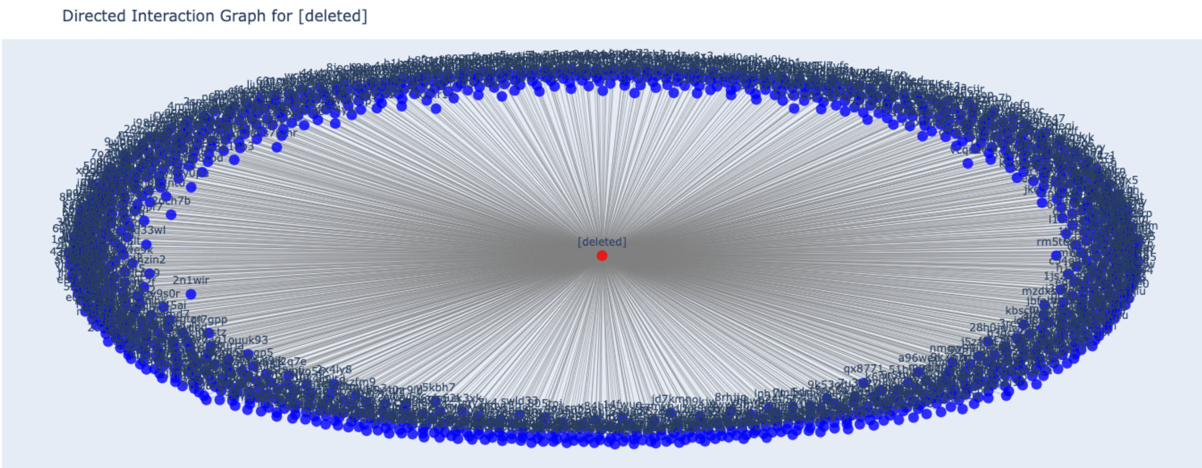




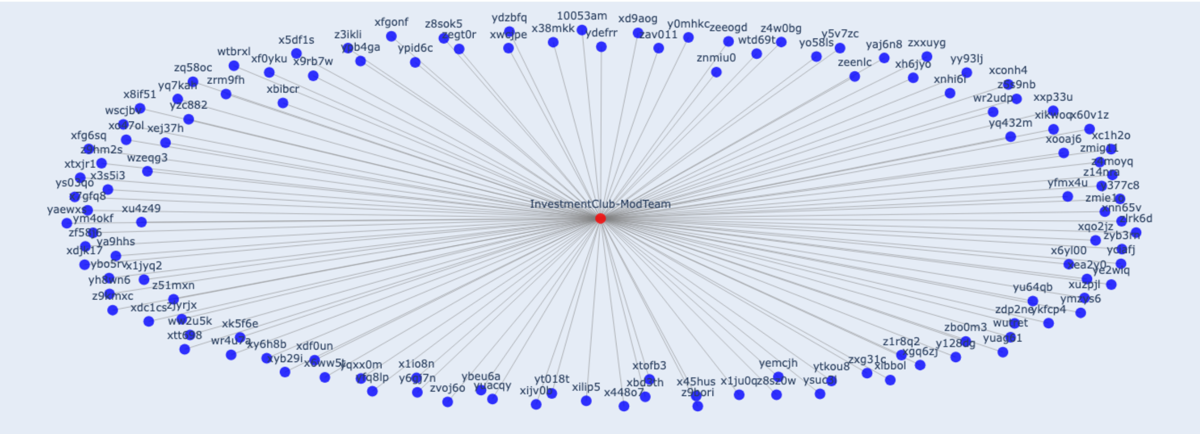
- **Directed Interaction Graph:**

  This visualization focuses on the interactions of the top 5 active users, showing directed edges where users reply to each other. The edges are directed from the commenter (parent) to the person being replied to (child).

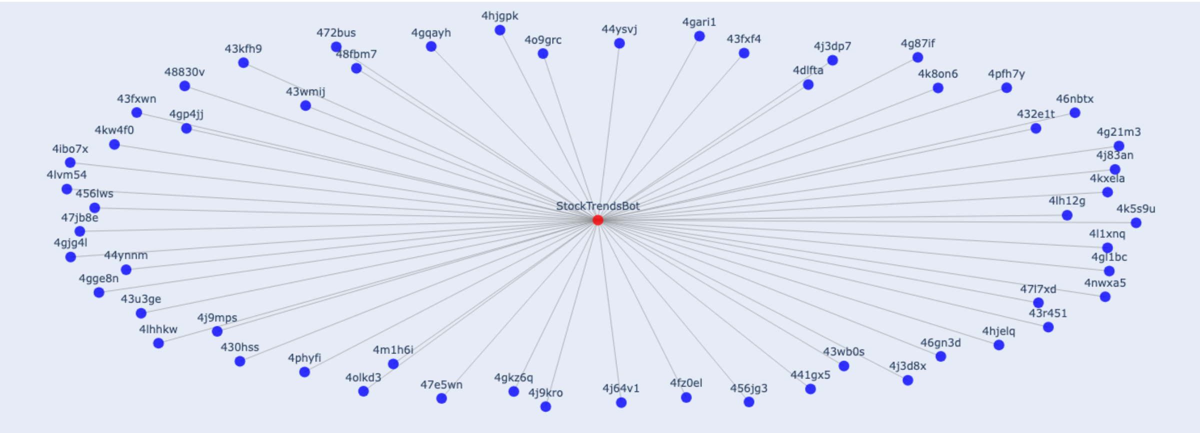




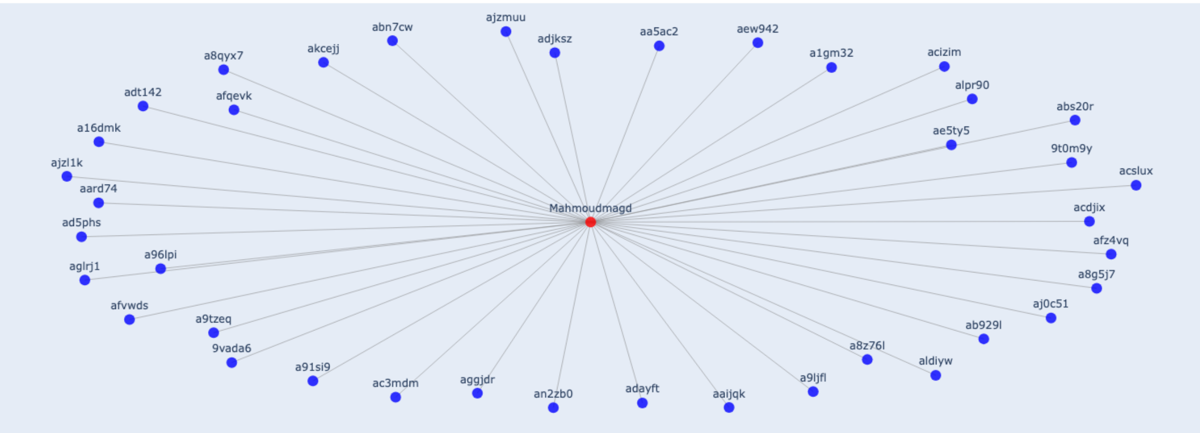
Directed Interaction Graph for InvestmentClub-ModTeam



Directed Interaction Graph for StockTrendsBot



Directed Interaction Graph for Mahmoudmagd



Pseudocode for Centrality Calculations:

<u>Calculate centrality for each node</u>

- Degree Centrality:
  - degree_centrality = nx.degree_centrality(G)
- Betweenness Centrality:
  - betweenness_centrality = nx.betweenness_centrality(G)

nodes.

- Closeness Centrality:

- closeness_centrality = nx.closeness_centrality(G)

<u>Analyse the results</u>

- Interpret which users are central to the graph based on the centrality measures.

3.3 Insights

From the metrics, **Zurevu** stands out as a central figure in the network, followed by **[deleted]**, with high **betweenness**, **closeness**, and **degree centrality**, indicating they are crucial for maintaining the flow of information and connecting various subgroups while the other users play important roles but to a lesser extent. These users play important roles in bridging groups and ensuring the smooth flow of interactions within the community.

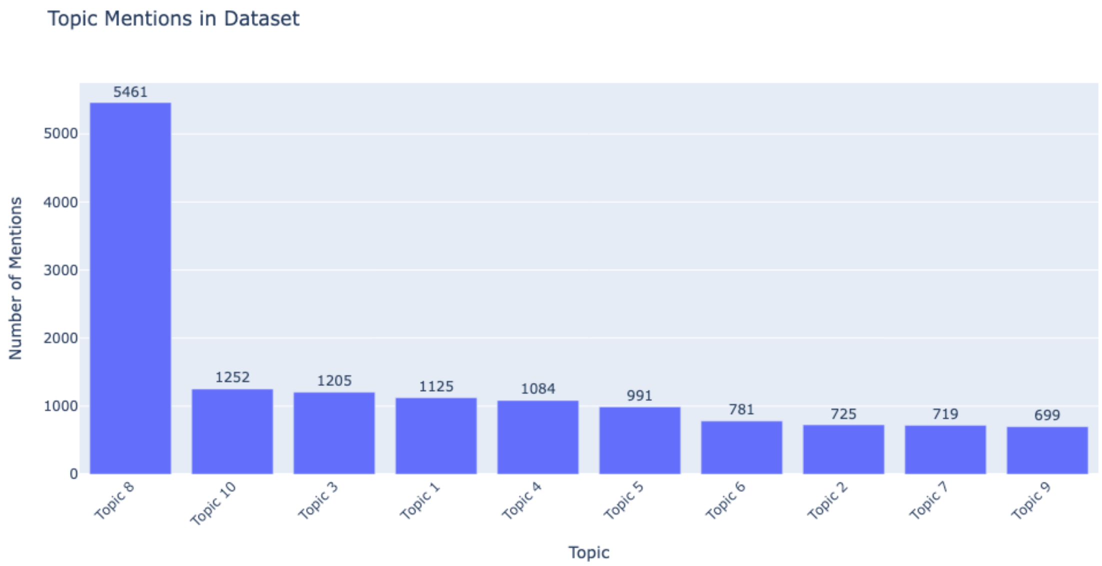
4. Super User Analysis

In this analysis, the focus was on understanding the role of super users in the community by using network analysis. The primary research question is to explore **how central super users are to the cohesiveness and functioning of the community**, particularly focusing on discussions involving Warren Buffett (a prominent topic within the community).

4.1. Data Preprocessing and Topic Modeling

To start the analysis, the text data from the subreddit was preprocessed by removing stopwords, lemmatizing, and cleaning the text. Then, **TF-IDF** vectorization was used to convert the cleaned text into a numerical matrix, followed by **UMAP** for dimensionality reduction. Additionally, **Latent Dirichlet Allocation (LDA)** was applied to identify the underlying topics discussed in the community. Below is a snapshot of a dataframe displaying the top keywords associated with each topic, followed by a bar chart that visualises the frequency of these topics.

topic_label	topic_name	row_count
Topic 8	stock, market, company, buy, investment, like, money, year, time, investor	5461
Topic 10	stock, fund, buy, invest, amp, investment, portfolio, 500, index, etfs	1252
Topic 3	buffett, warren, warren buffett, remove, stock, spam, market, berkshire, ticker report, buy	1205
Topic 1	bitcoin, crypto, referral, buy, code, invest, referral code, remove, stock, investment	1125
Topic 4	post, thank, quality, penny, penny stock, topic, sub, relevant, sub value, future	1084
Topic 5	delete, com, www, https, pitch, reddit, amp, http, remove, https www	991
Topic 6	beginner, advice, remove, mean, look advice, mean beginner, beginner look, beginner beginner, remove mean, comment remove	781
Topic 2	stock portfolio, portfolio, guru, stock, post, remove, competitor, portfolio competitor, post stock, specific stock	725
Topic 7	past, gold, musk, reit, elon, elon musk, year, delete, invest, rich	719
Topic 9	cryptocurrencie, cryptocurrencie topic, topic subreddit, subreddit allow, subreddit, remove, remove cryptocurrencie, allow, topic, analysis	699



Pseudocode for Topic Modeling:

<u>Text Data Preprocessing:</u>
For each comment:
- Remove stopwords
- Lemmatize the text
- Clean the text

<u>TF-IDF Vectorization:</u>
- vectorizer = TfidfVectorizer(max_features=5000, stop_words=list(custom_stopwords), ngram_range=(1, 2))
- X_tfidf = vectorizer.fit_transform(merged_data["cleaned_text"])

<u>Dimensionality Reduction (UMAP):</u>
- umap_model = UMAP(n_neighbors=20, n_components=10, min_dist=0.0, metric="cosine")

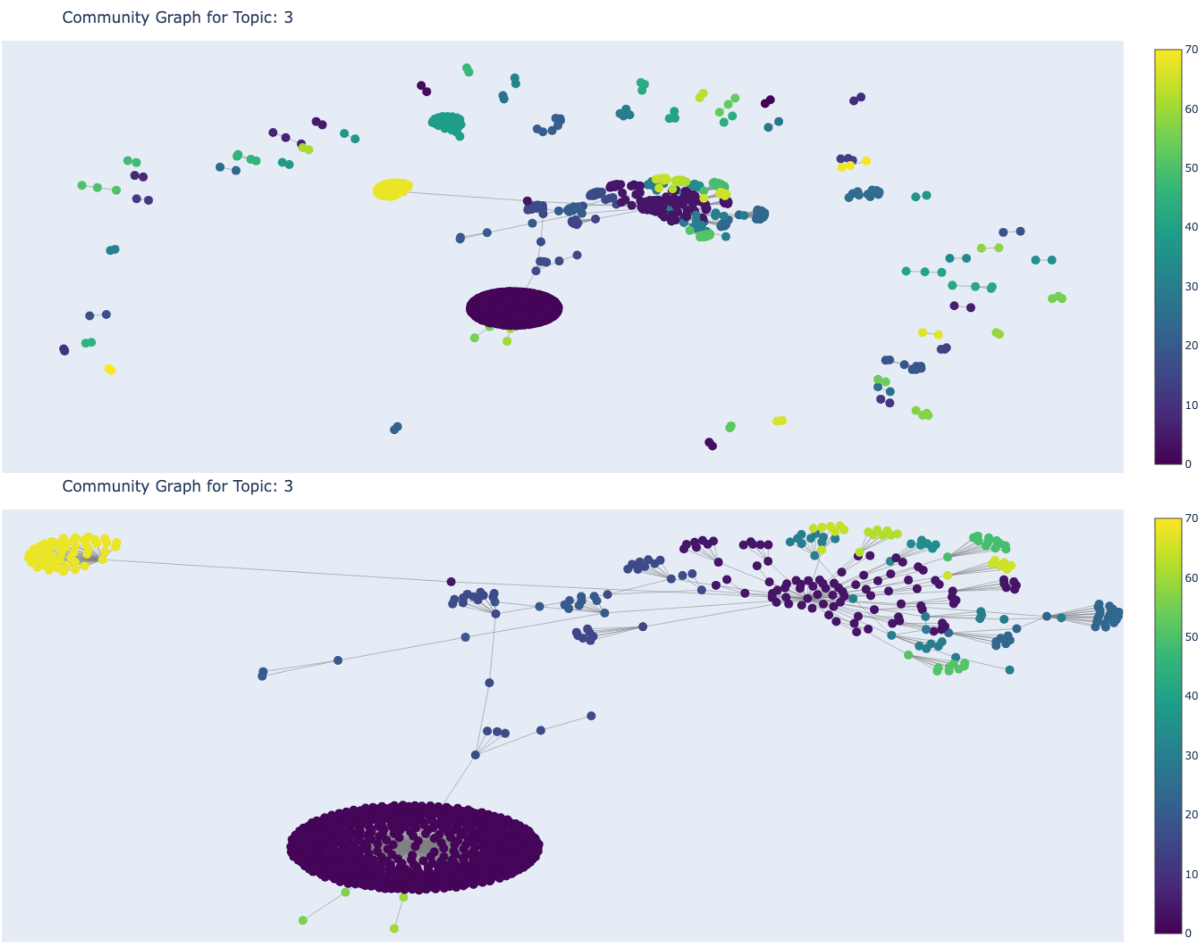
<u>Latent Dirichlet Allocation (LDA):</u>
- lda = LatentDirichletAllocation(n_components=10, random_state=42)
- lda.fit(X_tfidf)

<u>Visualize Topics:</u>
- Display top keywords for each topic from lda_model.components_
- Plot a bar chart showing the frequency of topics

4.2. Community Graph Construction for Topic 3: Warren Buffett Discussions

I used the **Louvain community detection algorithm** to build a community graph for each topic using user interactions (i.e., comment replies) as edges between users (i.e., nodes). The graph helps visualise the relationships between users and help us identify whether super users play a central role in the overall cohesiveness of the community.

Upon examining the results, I focused on the keywords related to the third most frequently discussed topic, which centered around Warren Buffett as it emerged as a more concrete and defined discussion theme having a well-connected network. Since Warren Buffett is a widely respected figure in investing, conversations around him indicate deeper engagement and knowledge exchange among users.



Community Graph for Warren Buffet Related Discussions

Pseudocode for Community Graph:

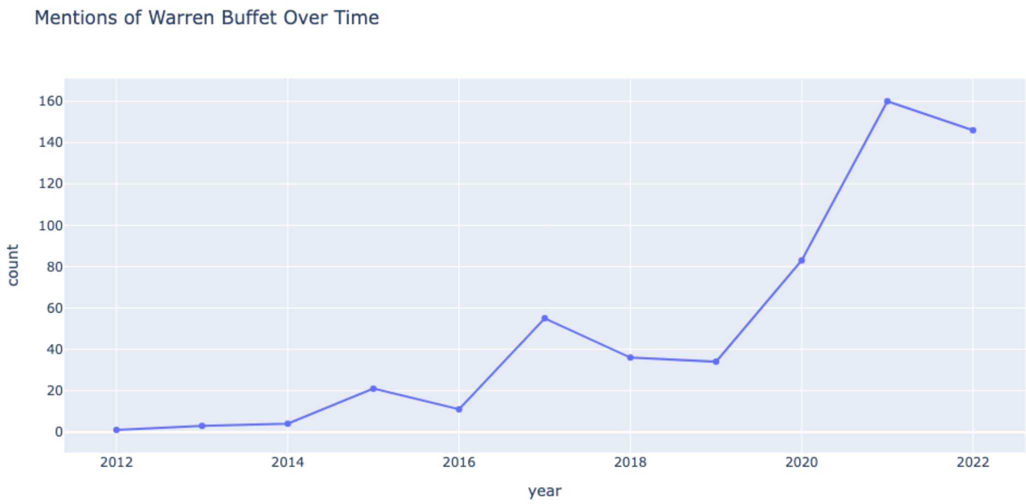
<u>Build Network Graph:</u>
Initialize the graph: G = nx.Graph()
For each user interaction (comment reply relationship):

- Add user nodes: G.add_node(user)
- Add edges between interacting users: G.add_edge(user, reply_to)

<u>Apply Louvain Community Detection:</u>
community = community_louvain.best_partition(G)

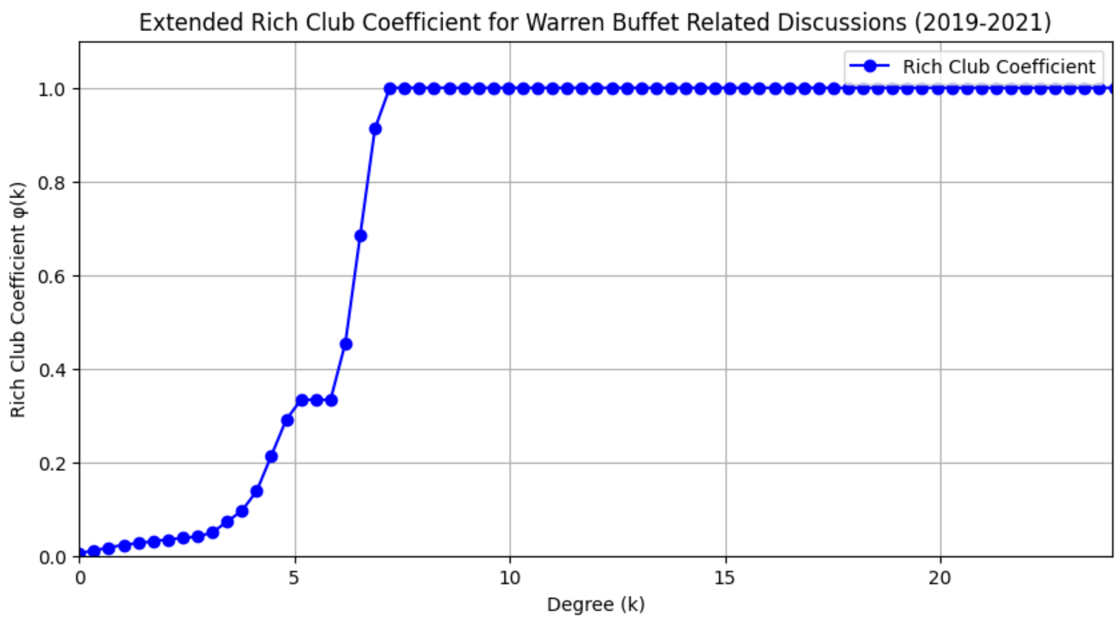
4.3. Topic Frequency Analysis Over Time

To study the evolution of discussions related to Warren Buffett, a time series analysis was conducted by counting the frequency of mentions over the years. The graph of Warren Buffett-related discussions over time revealed a peak in discussions starting from 2019.



4.4. Rich Club Coefficient

The **Rich Club Coefficient (RCC)** was computed to measure how the super users in the community contribute to network cohesion. The RCC quantifies the tendency for users with a higher number of interactions to form tightly connected clusters. I focused on the years 2019-2021 for the analysis when there was a surge in Warren Buffett-related discussions.



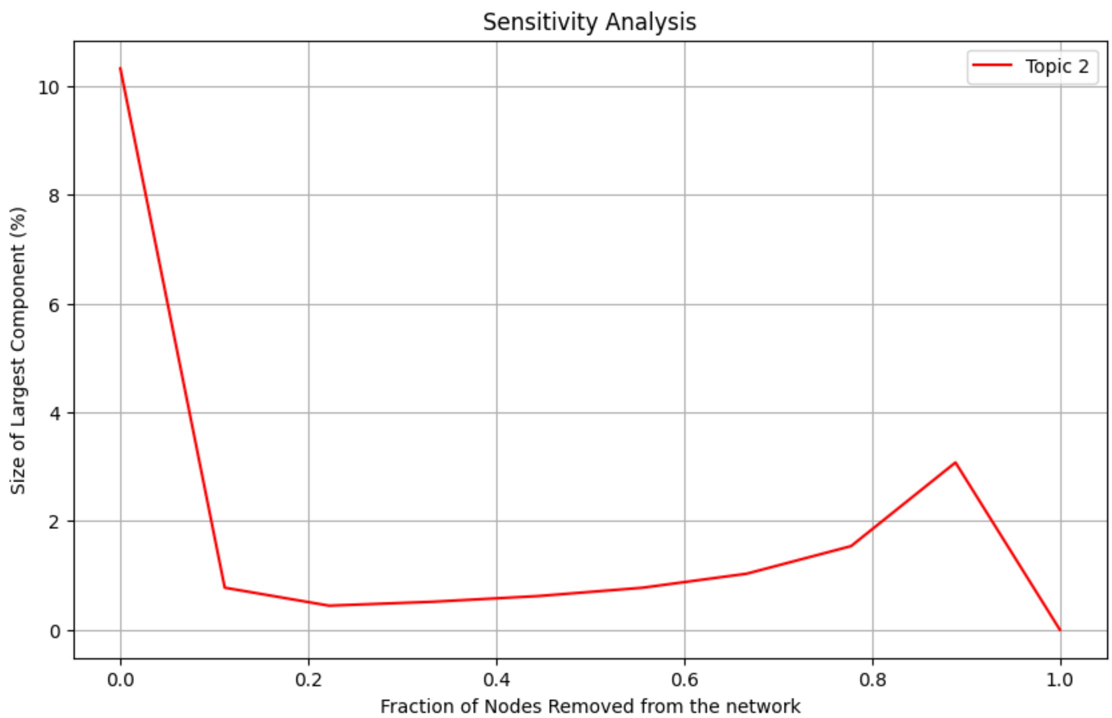
The plot of the rich club coefficient revealed that the degree of connectivity among highly connected users significantly increased, with coefficients reaching a value of 1 for users with degrees 7 and above indicating tightly-knit subgroups of super users. This suggests that super users in these discussions form a cohesive and exclusive network that is critical to the overall structure of the community.

Pseudocode:

rich_club = nx.rich_club_coefficient(G_warren, normalized=False)

4.5. Sensitivity Analysis

To assess the **resilience** of discussions on Warren Buffett, we conducted a **sensitivity analysis** by progressively removing high-degree users and observing the effect on the largest connected component.



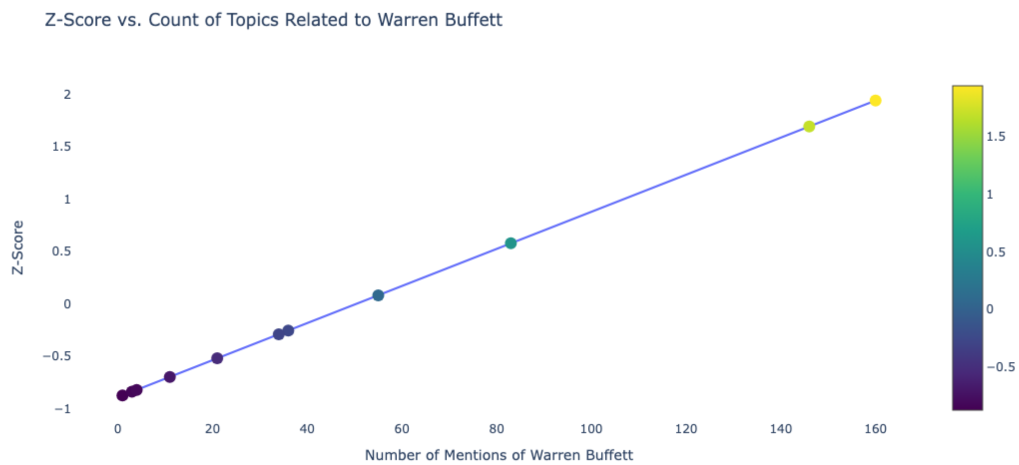
The sensitivity analysis revealed that initially, 10% of the nodes were part of the largest connected component. After removing just 10% of key users, this dropped sharply to 1%, highlighting the critical role that super users play in maintaining network cohesion.

These results suggest that Warren Buffett-related discussions are highly centralized, with a few dominant users driving the majority of engagement. If these key users were to leave, the structure of the discussion could fragment or weaken and hinder the dissemination of knowledge within financial communities.

4.6. Z-Score

The **Z-score** is a statistical measurement that tells you how many standard deviations a particular data point is from the mean (average) of a dataset. Initially, mentions of Warren Buffett were relatively consistent, with a few years exhibiting slightly fewer mentions, as indicated by negative Z-scores.

However, the analysis also highlights years with exceptionally **high Z-scores** (1.939 and 1.691), indicating a noticeable increase in mentions. These spikes suggest that specific events or news related to Warren Buffett may have significantly amplified his visibility in financial communities, driving more engagement and discussion around his views and activities. This implies that Warren Buffett's influence on financial conversations is not consistent year-round but rather driven by key moments that capture public and community attention.



Pseudocode:

z_score = (year_frequency - mean_frequency) / standard_deviation

4.7 Insights

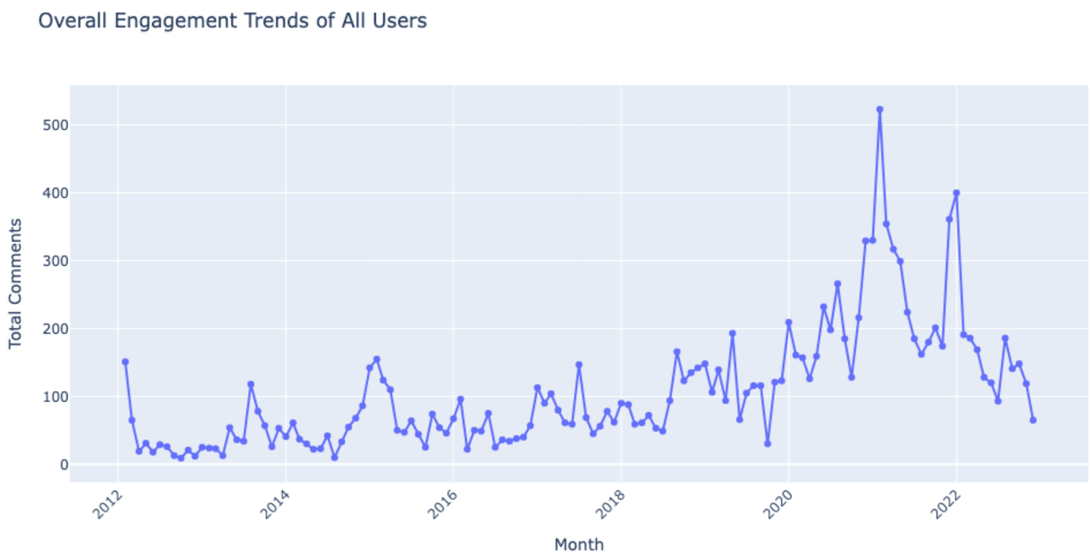
This analysis has highlighted the crucial role super users play in maintaining the cohesion and functionality of the community. By leveraging network analysis techniques such as centrality and the Rich Club Coefficient, it was evident that these users, through their high levels of engagement form the backbone of community interactions, especially in discussions surrounding Warren Buffett.

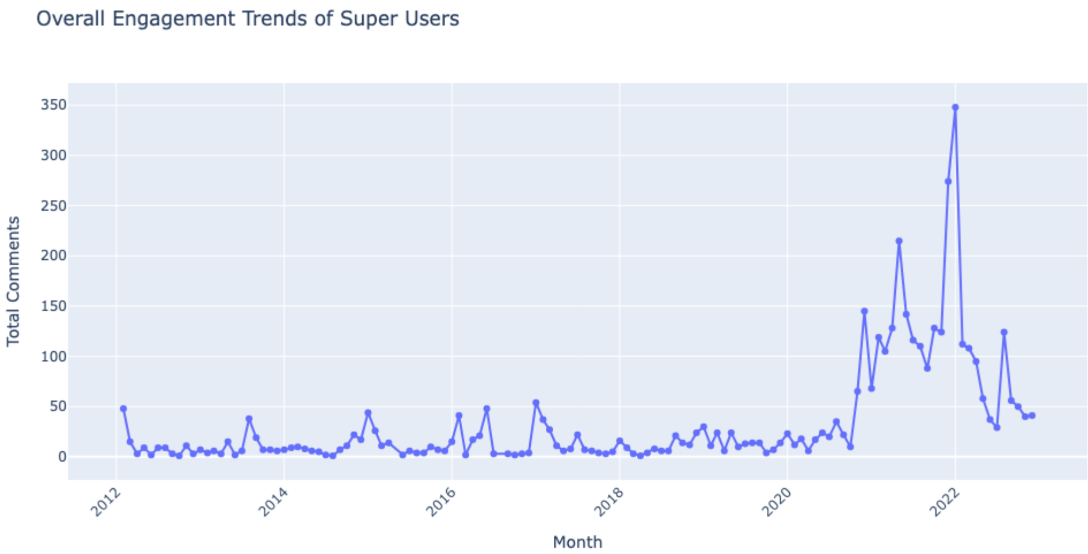
5. Temporal Dynamics and Key Topic Identification

We wanted to investigate whether there were any trends or spikes in engagement over time, and to do this we first analysed the **monthly engagement trends** of all users and super users over the years.

5.1 Monthly Engagement Trends

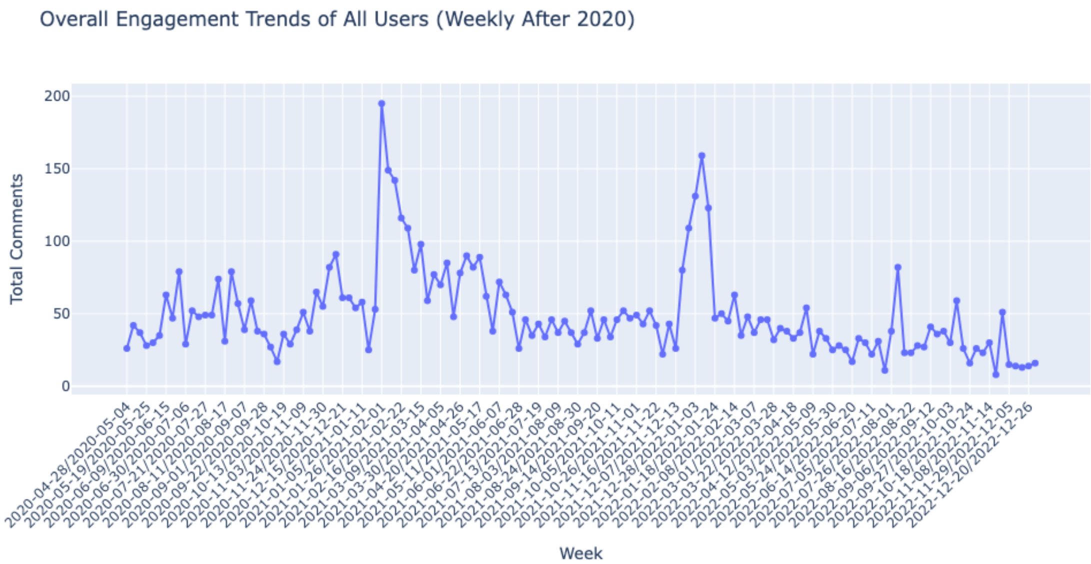
By aggregating the number of comments per month, we observed clear fluctuations in engagement. The visualizations of monthly trends over the years revealed a **significant spike in activity after 2020**. Notably, super users showed a parallel rise in engagement, indicating their strong role in driving and sustaining interactions within the subreddit.

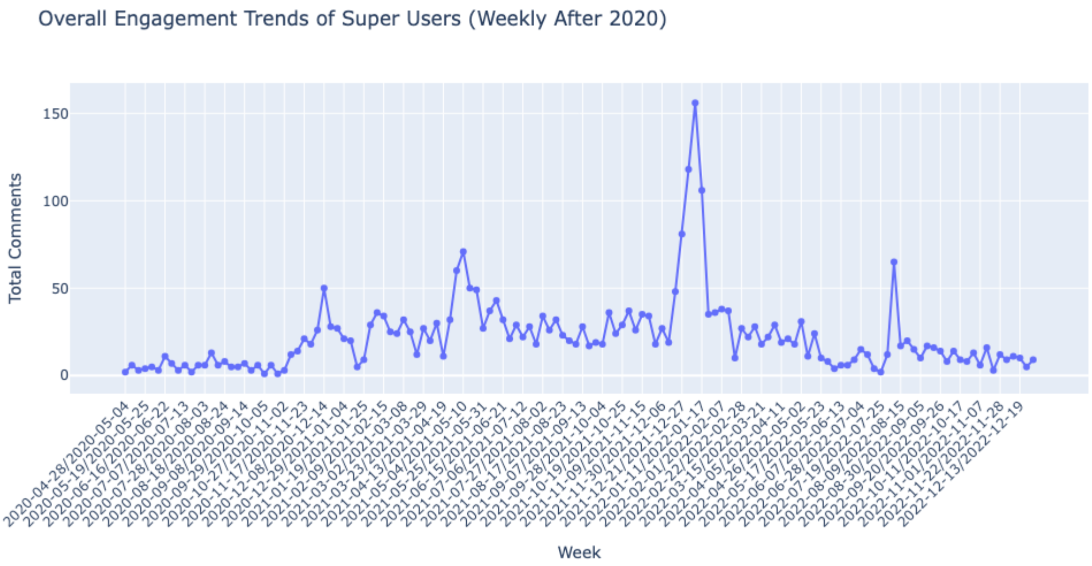




5.2 Weekly Engagement Trends After 2020

Given the noticeable increase in activity post-2020, we conducted a more granular analysis by breaking down engagement on a weekly basis. This allowed us to pinpoint specific periods of heightened activity that stood out against the general trend.





Through this analysis, we identified two distinct peak periods where engagement was significantly higher than usual:

- **January 26 – February 1, 2021**
- **January 4 – January 10, 2022**

These periods warranted further investigation to understand the driving factors behind these surges. To explore this, we manually analysed the comments to identify key discussion themes and topics during these spikes.

5.3 Insights

The first spike in discussions was sparked by the **GameStop (GME)** short squeeze where retail investors pushed the stock price up forcing hedge funds to cover their short positions at a loss. This led to debates about market manipulation**,** especially regarding brokerage platforms like Robinhood which restricted trading during the surge.

The second spike continued the conversation on short squeezes, focusing on stocks like Nokia (NOK) and AMC, and expanded to include broader investment strategies. Discussions also centered around **Warren Buffett's influence**, as users compared his long-term investment approach to the tactics of retail investors and institutional players.

Conclusion

This study of the dataset reveals the critical role super users play in the cohesiveness and functioning of the community. We found that super users significantly influence the flow of information, shape discussions, and maintain a strong presence in the community. Their high degree of centrality measured through metrics such as betweenness centrality, closeness centrality and degree centrality indicate that these users are central to the overall engagement and stability of the community.

The topic modeling and sensitivity analysis provided further insights into how discussions, particularly those surrounding key figures like Warren Buffett are highly influenced by a small group of dominant users. These super users are critical to the success of key discussions and can even determine the longevity and flow of interactions within the subreddit. The analysis of the Rich Club Coefficient and Z-scores further reinforced the idea that discussions around prominent figures, such as Buffett are not evenly distributed and the community's engagement is heavily driven by certain key moments, especially during market shifts.

Additionally, the investigation into the temporal dynamics of user interactions revealed a noticeable spike in user activity after 2020, particularly around key financial events. This aligns with the findings that external events, such as the **GameStop (GME)** short squeeze and shifts in market trends, significantly influenced the volume of discussions and user engagement. This period saw increased participation with prominent discussions focusing on market manipulation, investment strategies and influential figures like **Warren Buffett**.

Ultimately, this research underscores the importance of super users in maintaining the vibrancy and cohesion of online communities. These insights are valuable for understanding how online communities can be nurtured and sustained especially in the context of financial forums where knowledge sharing and user participation are key to the overall success of the platform.